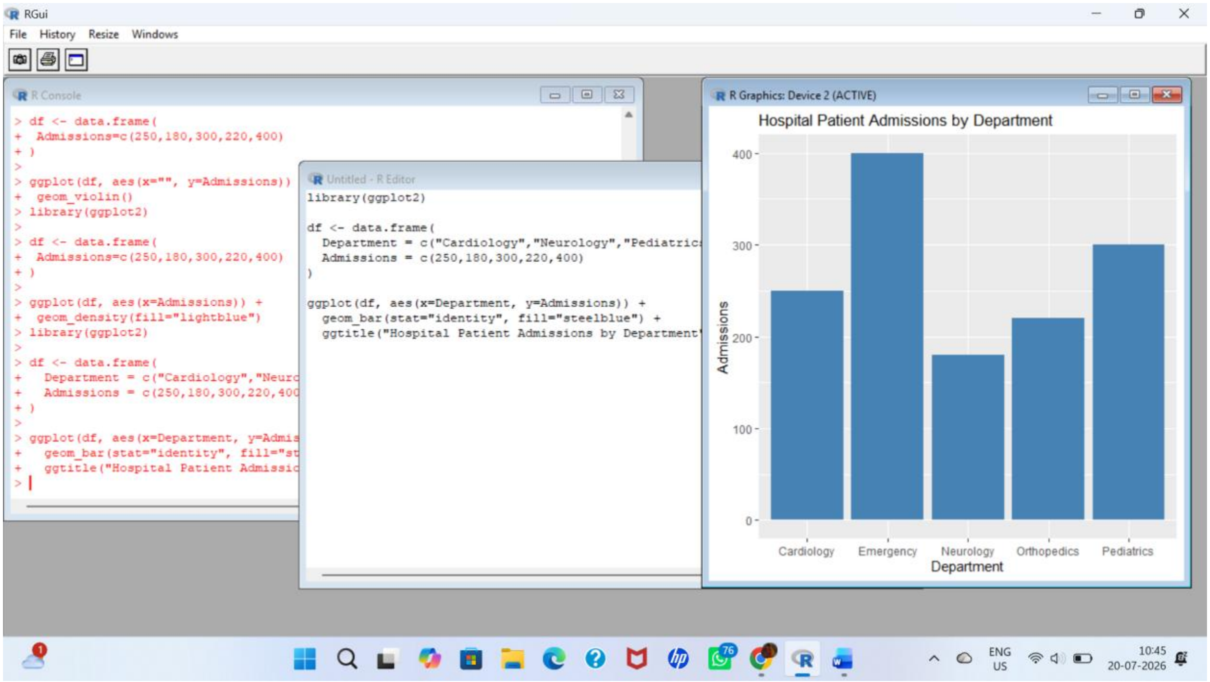
CLASS ACTIVITY – 20-07-2026

SUDHIREDDY DEEKSHITHA

Question 1 – Hospital Patient Admission Analysis

1. Bar Chart

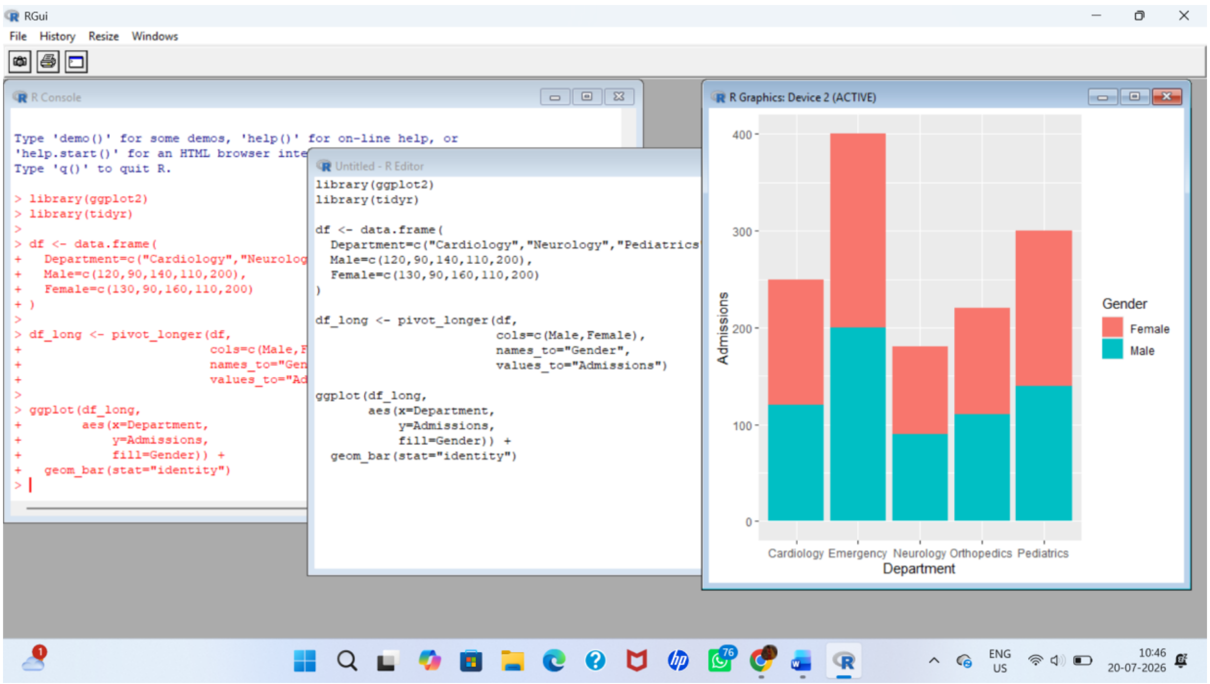
```
library(ggplot2)
df <- data.frame(
  Department = c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),
  Admissions = c(250,180,300,220,400)
)
ggplot(df, aes(x=Department, y=Admissions)) +
  geom_bar(stat="identity", fill="steelblue") +
  ggtitle("Hospital Patient Admissions by Department")
```



2. Stacked Bar Chart

```
library(ggplot2)
library(tidyr)
df <- data.frame(
  Department=c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),
  Male=c(120,90,140,110,200),
```

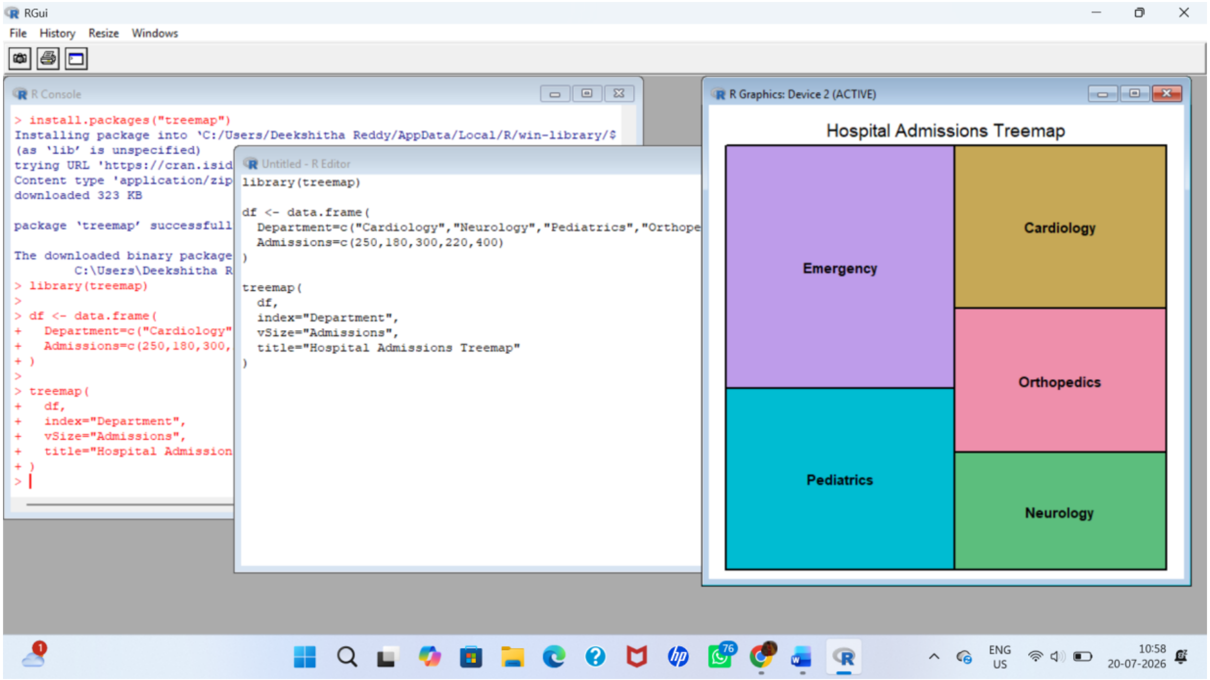
```
  Female=c(130,90,160,110,200)
)
df_long <- pivot_longer(df,
                        cols=c(Male,Female),
                        names_to="Gender",
                        values_to="Admissions")
ggplot(df_long,
       aes(x=Department,
           y=Admissions,
           fill=Gender)) +
  geom_bar(stat="identity")
```



3. Treemap

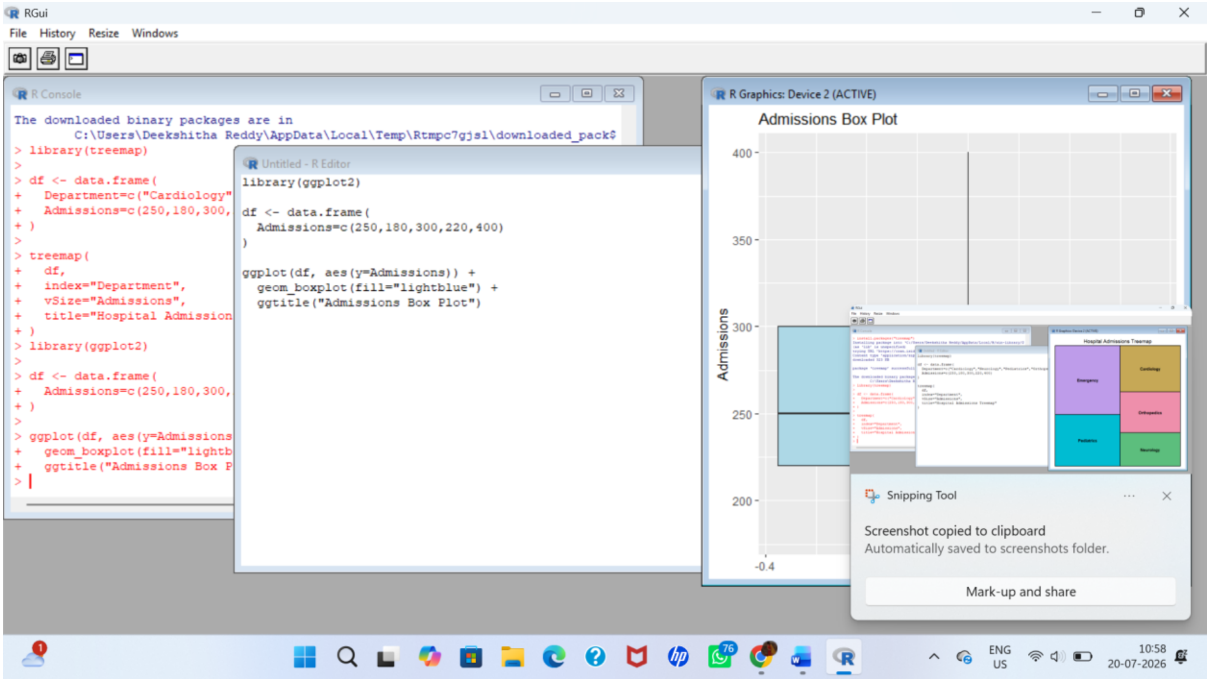
```
library(treemap)
df <- data.frame(
  Department=c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),
  Admissions=c(250,180,300,220,400)
)
```

```
treemap(
  df,
  index="Department",
  vSize="Admissions",
  title="Hospital Admissions Treemap"
)
```



4. Box Plot

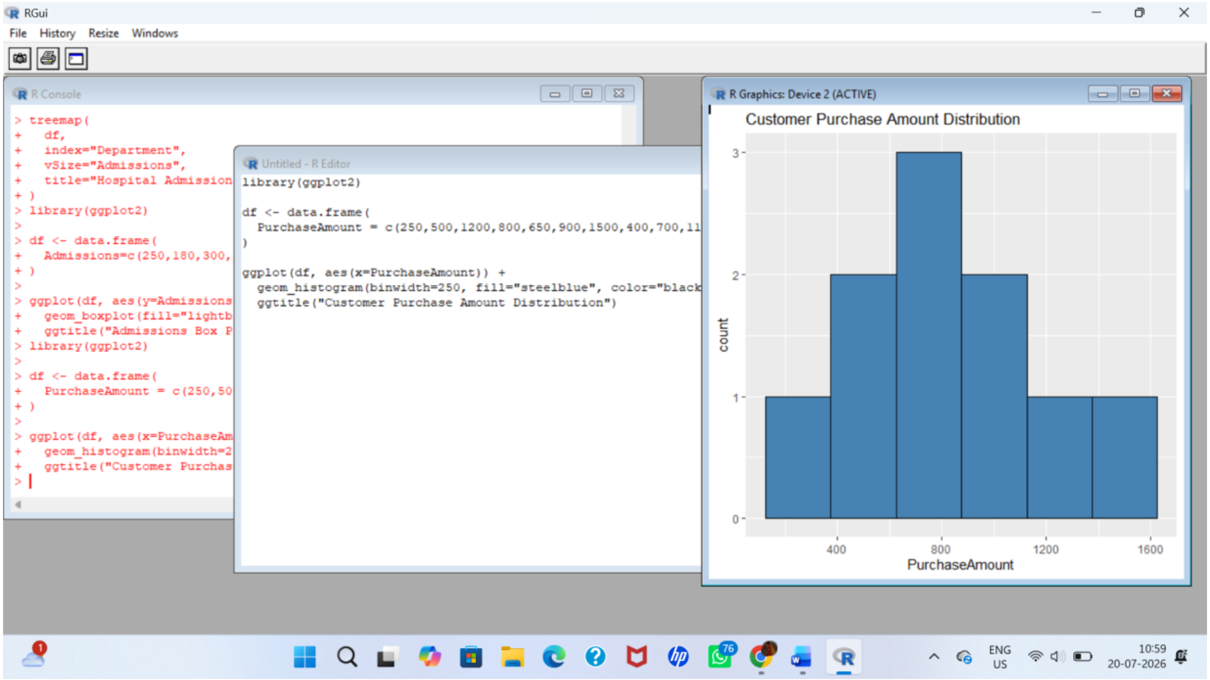
```
library(ggplot2)
df <- data.frame(
  Admissions=c(250,180,300,220,400)
)
ggplot(df, aes(y=Admissions)) +
  geom_boxplot(fill="lightblue") +
  ggtitle("Admissions Box Plot")v
```



Question 2 – E-Commerce Customer Spending Behaviour

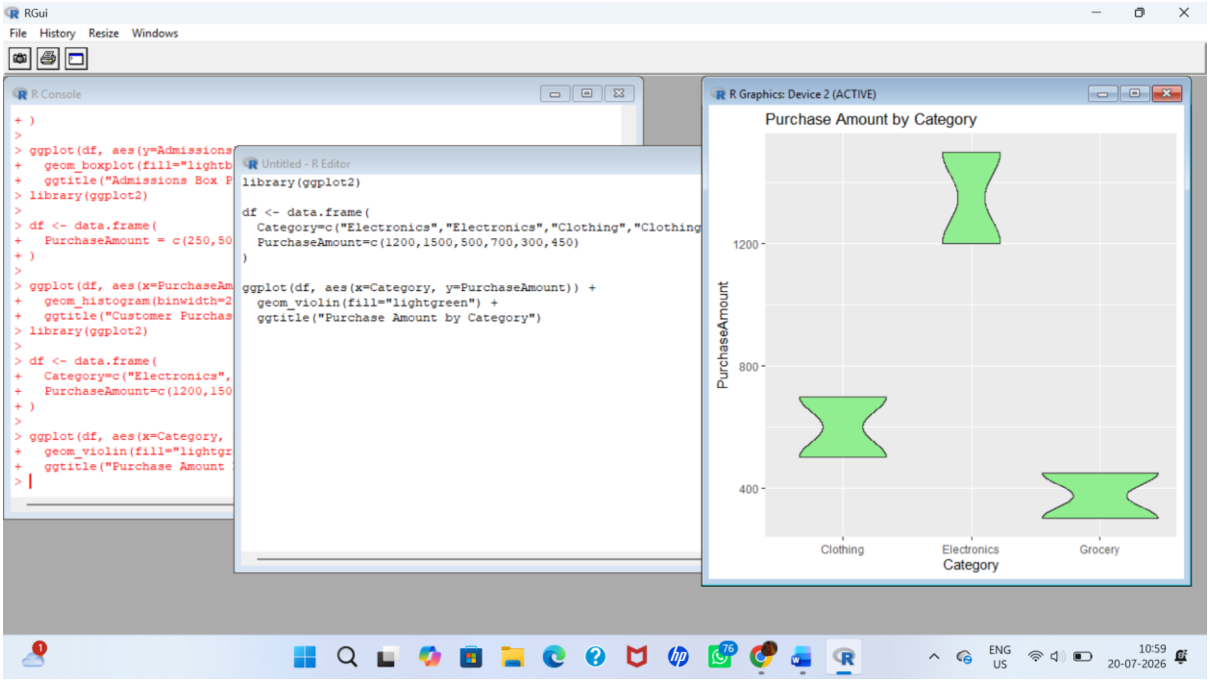
1. Histogram

```
library(ggplot2)

df <- data.frame(

  PurchaseAmount = c(250,500,1200,800,650,900,1500,400,700,1100)

)

ggplot(df, aes(x=PurchaseAmount)) +

  geom_histogram(binwidth=250, fill="steelblue", color="black") +

  ggtitle("Customer Purchase Amount Distribution")
```



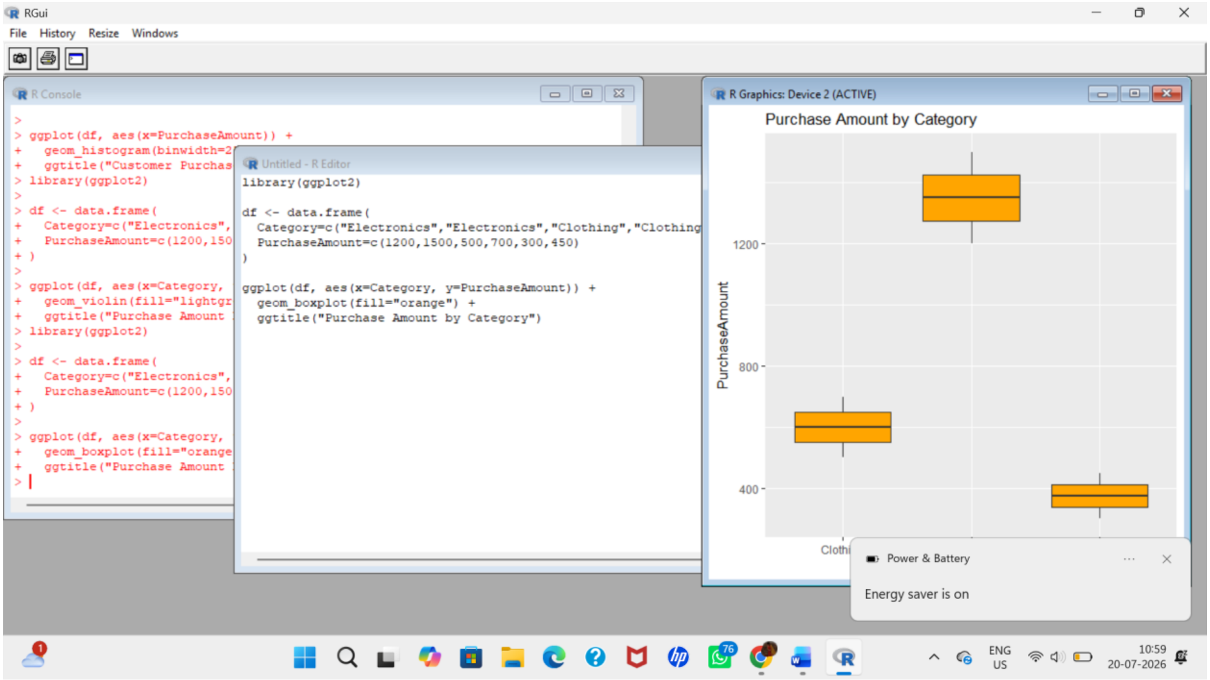
2. Violin Plot

```
library(ggplot2)

df <- data.frame(
  Category=c("Electronics","Electronics","Clothing","Clothing","Grocery","Grocery"),
  PurchaseAmount=c(1200,1500,500,700,300,450)
)

ggplot(df, aes(x=Category, y=PurchaseAmount)) +
  geom_violin(fill="lightgreen") +
  ggtitle("Purchase Amount by Category")
```



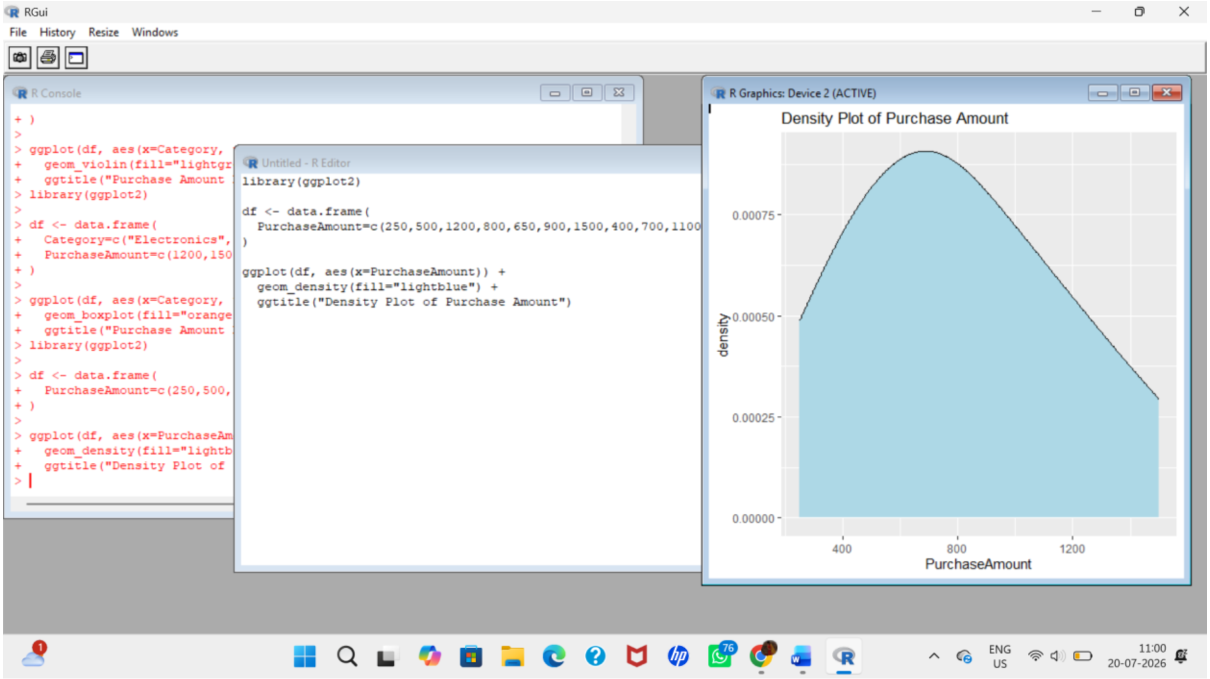
3. Box Plot

```
library(ggplot2)
df <- data.frame(
  Category=c("Electronics","Electronics","Clothing","Clothing","Grocery","Grocery"),
  PurchaseAmount=c(1200,1500,500,700,300,450)
)
ggplot(df, aes(x=Category, y=PurchaseAmount)) +
  geom_boxplot(fill="orange") +
  ggtitle("Purchase Amount by Category")
```



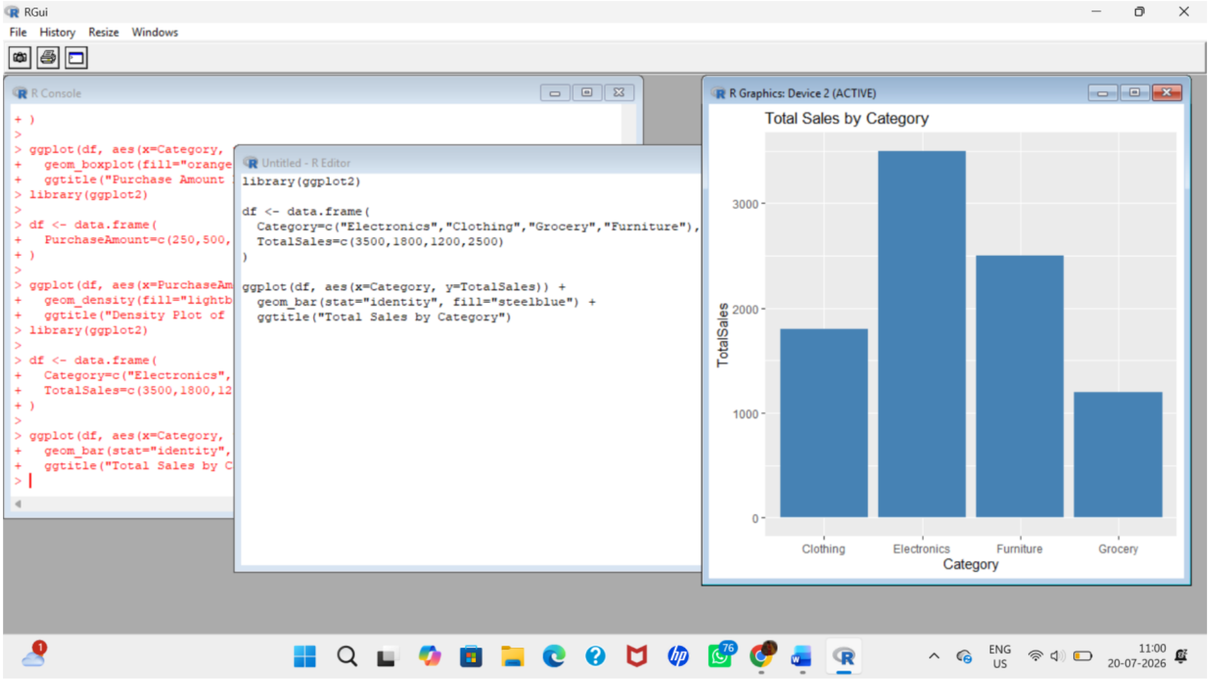
4. Density Plot

```
library(ggplot2)

df <- data.frame(
  PurchaseAmount=c(250,500,1200,800,650,900,1500,400,700,1100)
)

ggplot(df, aes(x=PurchaseAmount)) +
  geom_density(fill="lightblue") +
  ggtitle("Density Plot of Purchase Amount")
```



5. Bar Chart

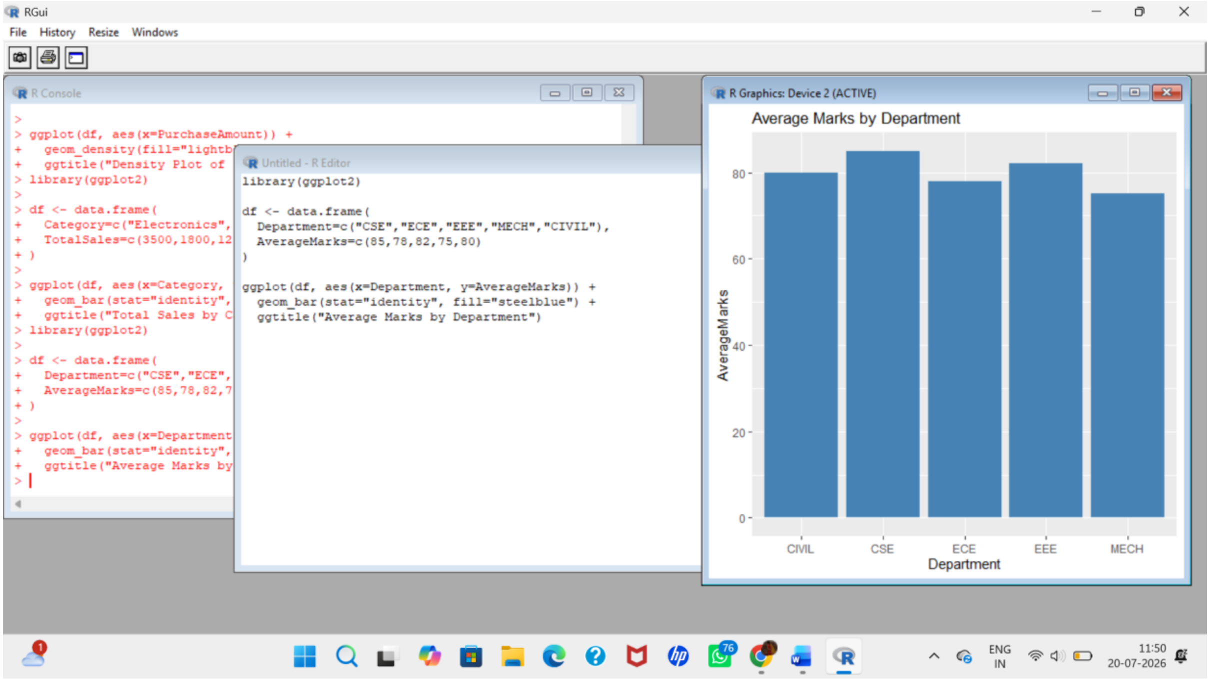
```
library(ggplot2)

df <- data.frame(
  Category=c("Electronics","Clothing","Grocery","Furniture"),
  TotalSales=c(3500,1800,1200,2500)
)

ggplot(df, aes(x=Category, y=TotalSales)) +
  geom_bar(stat="identity", fill="steelblue") +
  ggtitle("Total Sales by Category")
```



Question 3 – University Examination Performance Dashboard

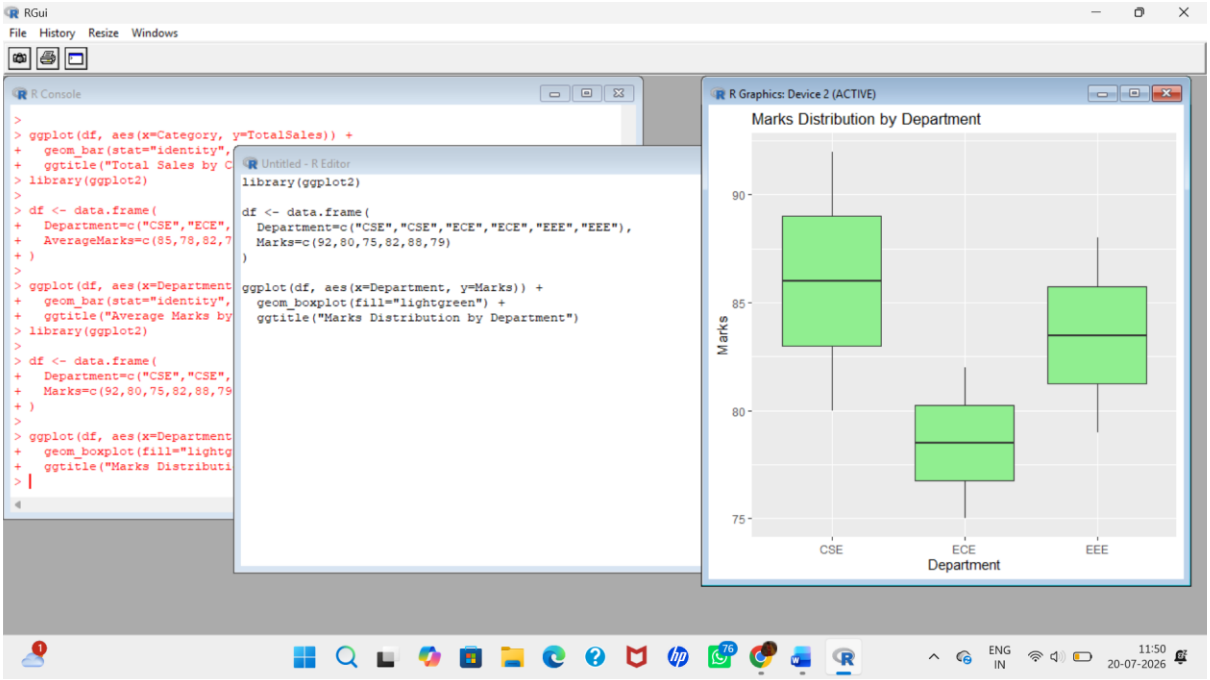
1. Bar Chart

```
library(ggplot2)

df <- data.frame(
  Department=c("CSE","ECE","EEE","MECH","CIVIL"),
  AverageMarks=c(85,78,82,75,80)
)

ggplot(df, aes(x=Department, y=AverageMarks)) +
  geom_bar(stat="identity", fill="steelblue") +
  ggtitle("Average Marks by Department")
```



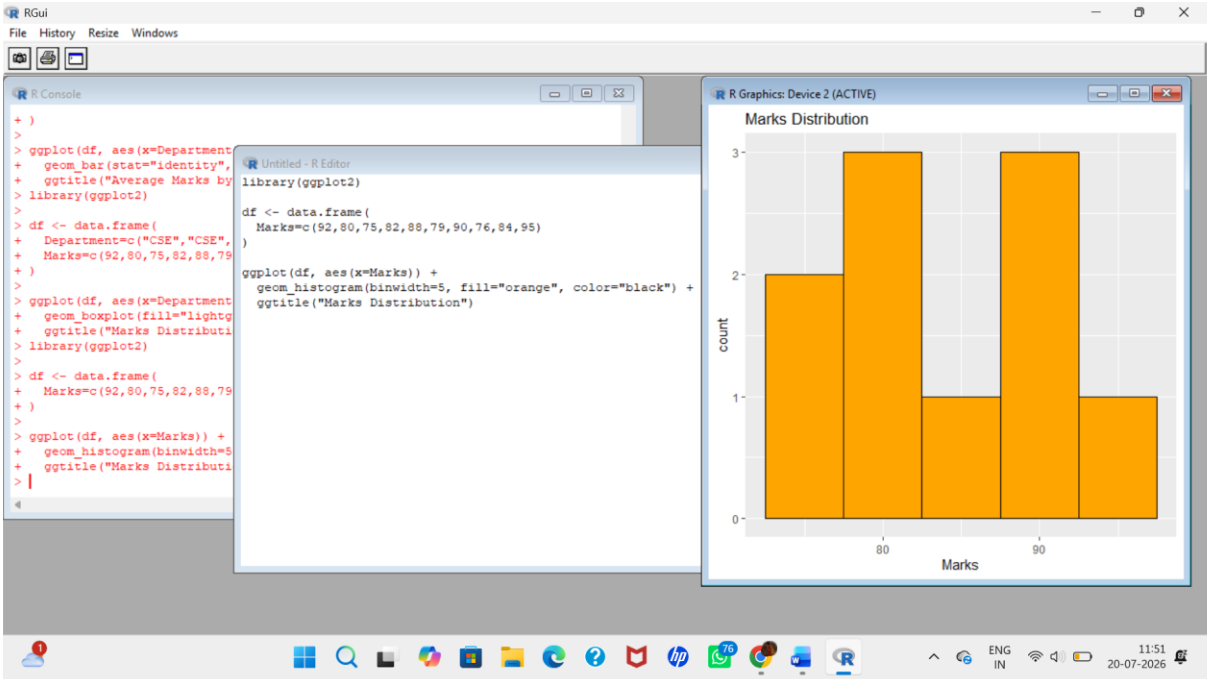
2. Box Plot

```
library(ggplot2)

df <- data.frame(
  Department=c("CSE","CSE","ECE","ECE","EEE","EEE"),
  Marks=c(92,80,75,82,88,79)
)

ggplot(df, aes(x=Department, y=Marks)) +
  geom_boxplot(fill="lightgreen") +
  ggtitle("Marks Distribution by Department")
```



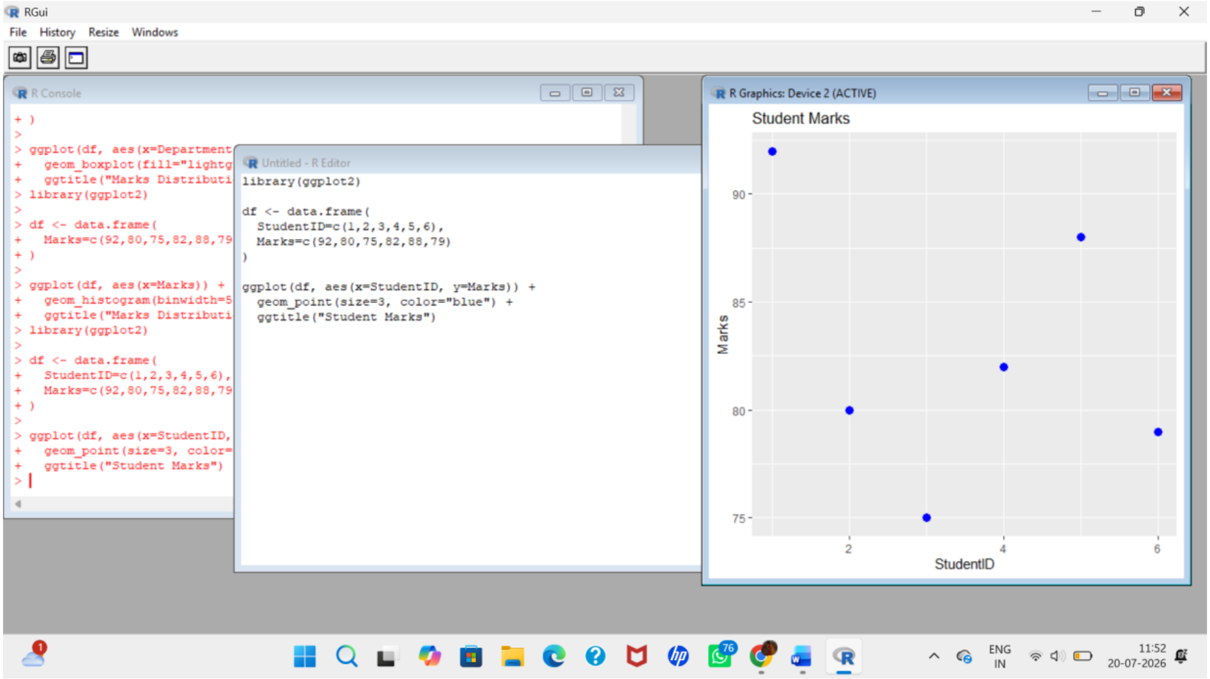
3. Histogram

```
library(ggplot2)

df <- data.frame(
  Marks=c(92,80,75,82,88,79,90,76,84,95)
)

ggplot(df, aes(x=Marks)) +
  geom_histogram(binwidth=5, fill="orange", color="black") +
  ggtitle("Marks Distribution")
```



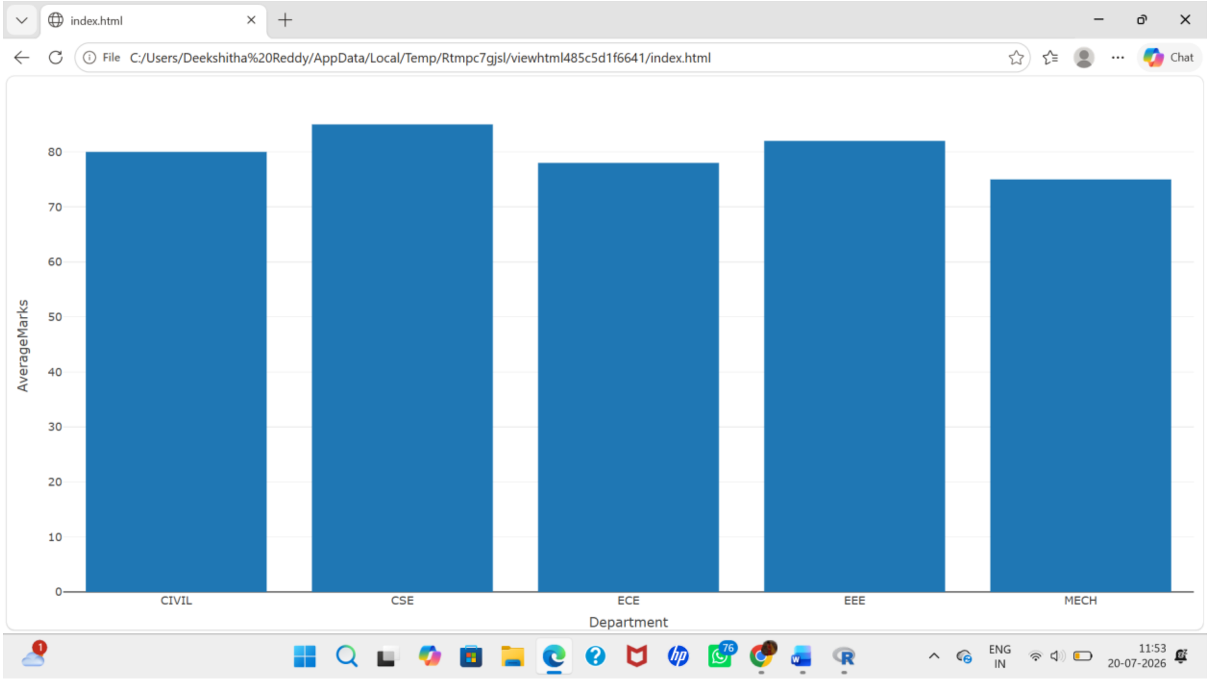
4. Scatter Plot

```
library(ggplot2)
df <- data.frame(
  StudentID=c(1,2,3,4,5,6),
  Marks=c(92,80,75,82,88,79)
)
ggplot(df, aes(x=StudentID, y=Marks)) +
  geom_point(size=3, color="blue") +
  ggtitle("Student Marks")
```



5. Interactive Bar Chart (Plotly)

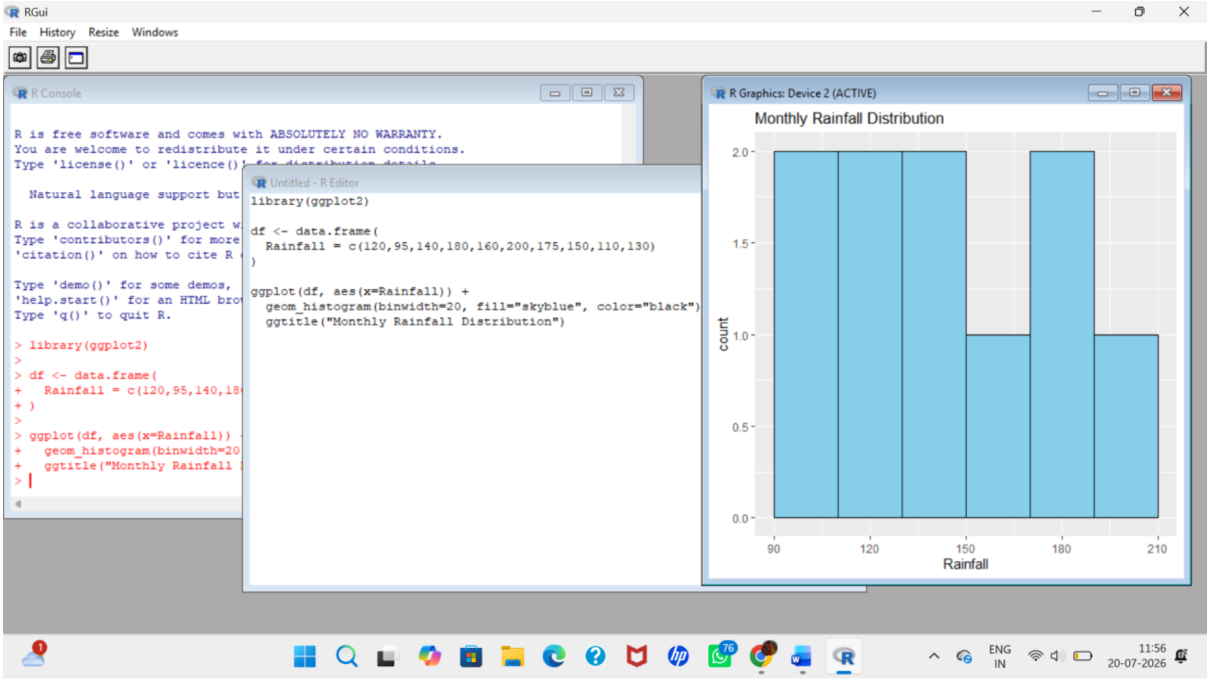
```
library(plotly)
df <- data.frame(
  Department=c("CSE","ECE","EEE","MECH","CIVIL"),
  AverageMarks=c(85,78,82,75,80)
)
plot_ly(
  df,
  x=~Department,
  y=~AverageMarks,
  type='bar'
)
```



Question 4 – Climate and Rainfall Distribution Study

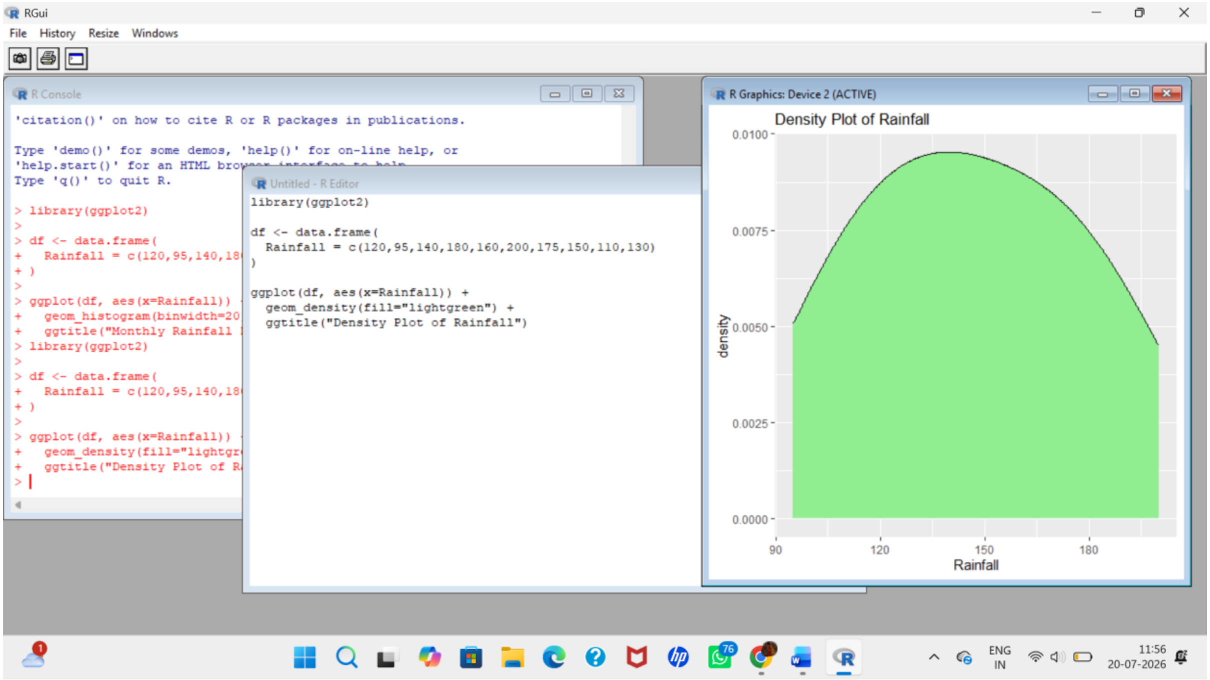
1. Histogram

```
library(ggplot2)
df <- data.frame(
  Rainfall = c(120,95,140,180,160,200,175,150,110,130)
)
ggplot(df, aes(x=Rainfall)) +
  geom_histogram(binwidth=20, fill="skyblue", color="black") +
  ggtitle("Monthly Rainfall Distribution")
```



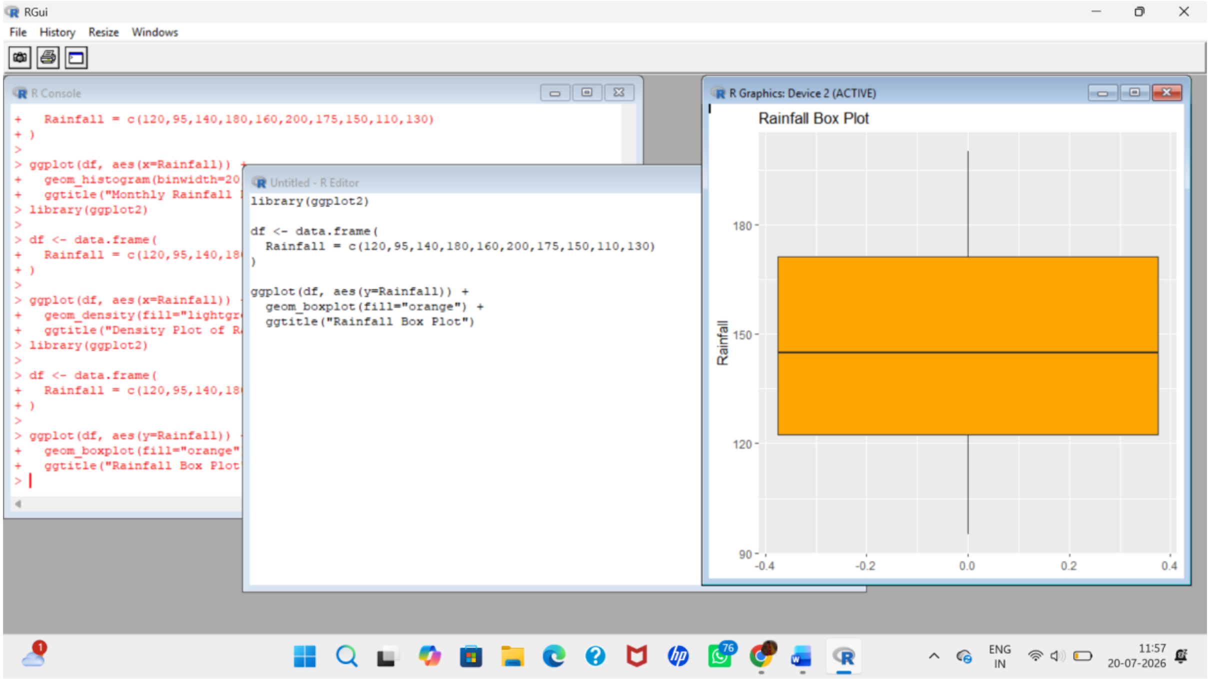
2. Density Plot

```
library(ggplot2)

df <- data.frame(

  Rainfall = c(120,95,140,180,160,200,175,150,110,130)

)

ggplot(df, aes(x=Rainfall)) +

  geom_density(fill="lightgreen") +

  ggtitle("Density Plot of Rainfall")
```



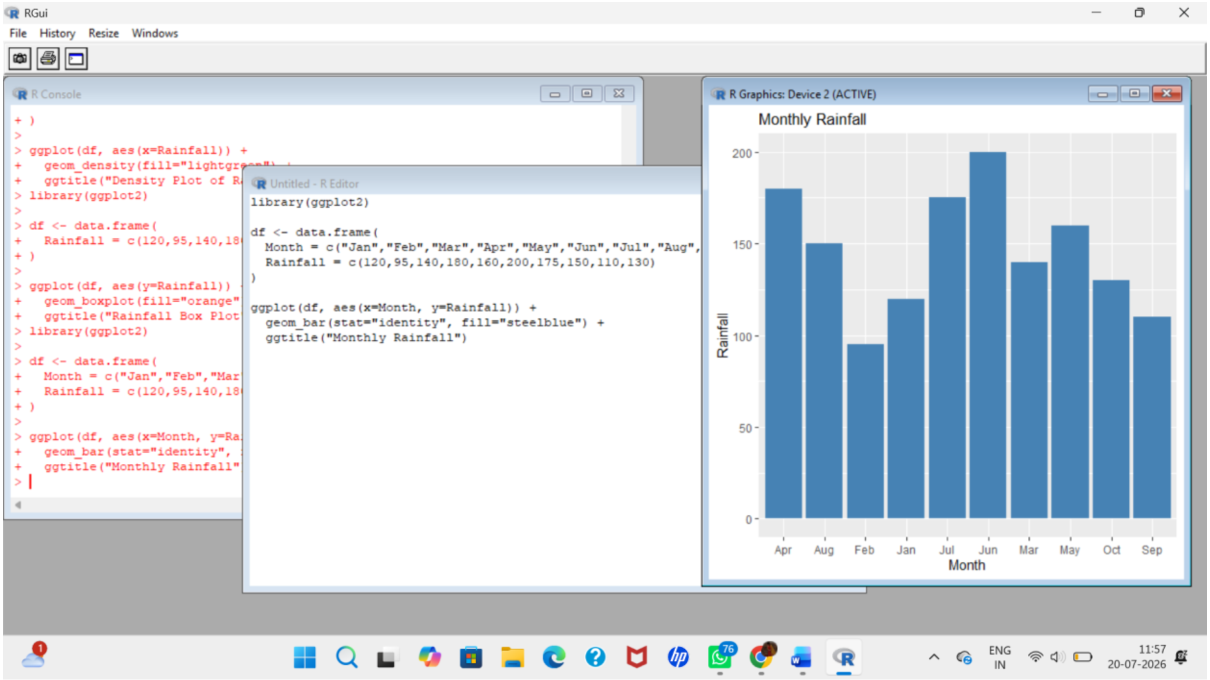
3. Box Plot

```
library(ggplot2)
df <- data.frame(
  Rainfall = c(120,95,140,180,160,200,175,150,110,130)
)
ggplot(df, aes(y=Rainfall)) +
  geom_boxplot(fill="orange") +
  ggtitle("Rainfall Box Plot")
```



4. Bar Chart

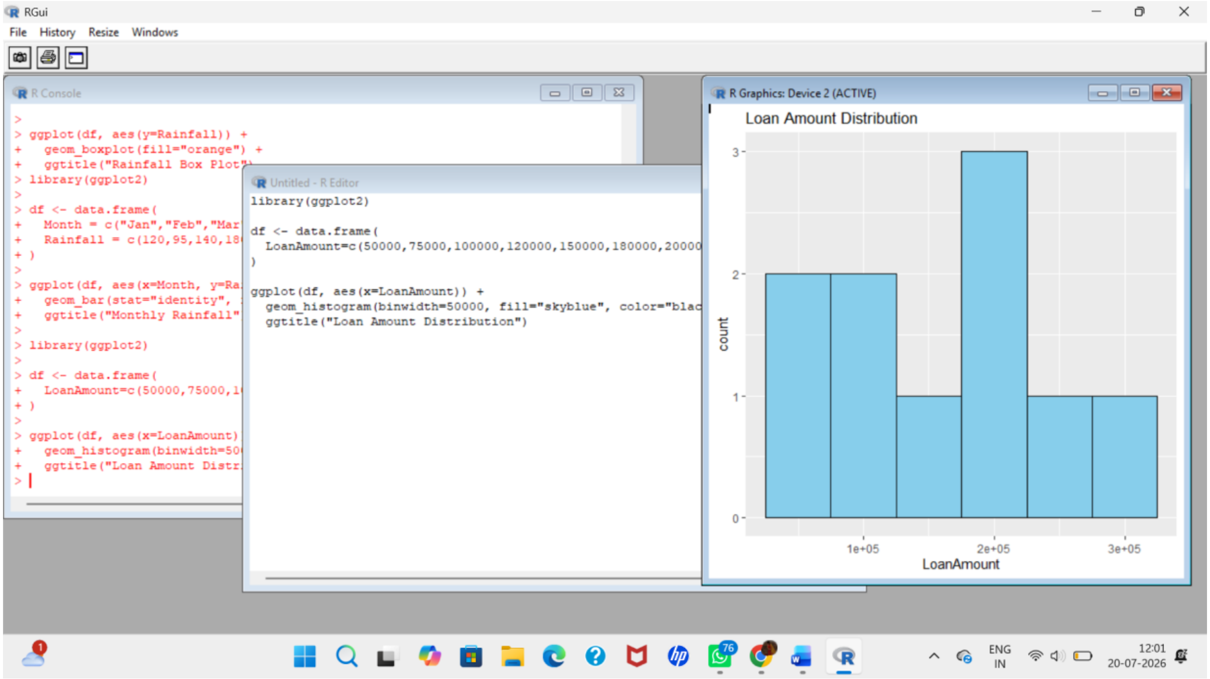
```
library(ggplot2)
df <- data.frame(
  Month = c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct"),
  Rainfall = c(120,95,140,180,160,200,175,150,110,130)
)
ggplot(df, aes(x=Month, y=Rainfall)) +
  geom_bar(stat="identity", fill="steelblue") +
  ggtitle("Monthly Rainfall")
```



Question 5 – Bank Loan Risk Assessment

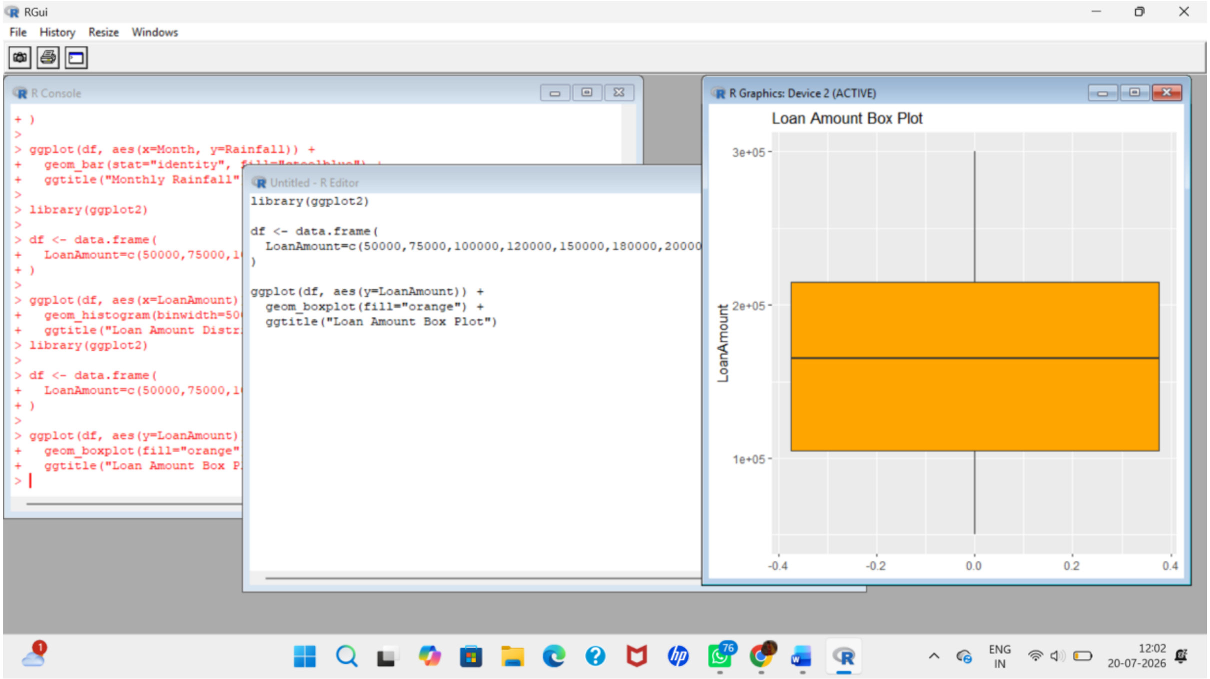
1. Histogram (Loan Amount Distribution)

```
library(ggplot2)

df <- data.frame(

LoanAmount=c(50000,75000,100000,120000,150000,180000,200000,220000,250000,300000)

)

ggplot(df, aes(x=LoanAmount)) +

  geom_histogram(binwidth=50000, fill="skyblue", color="black") +

  ggtitle("Loan Amount Distribution")
```



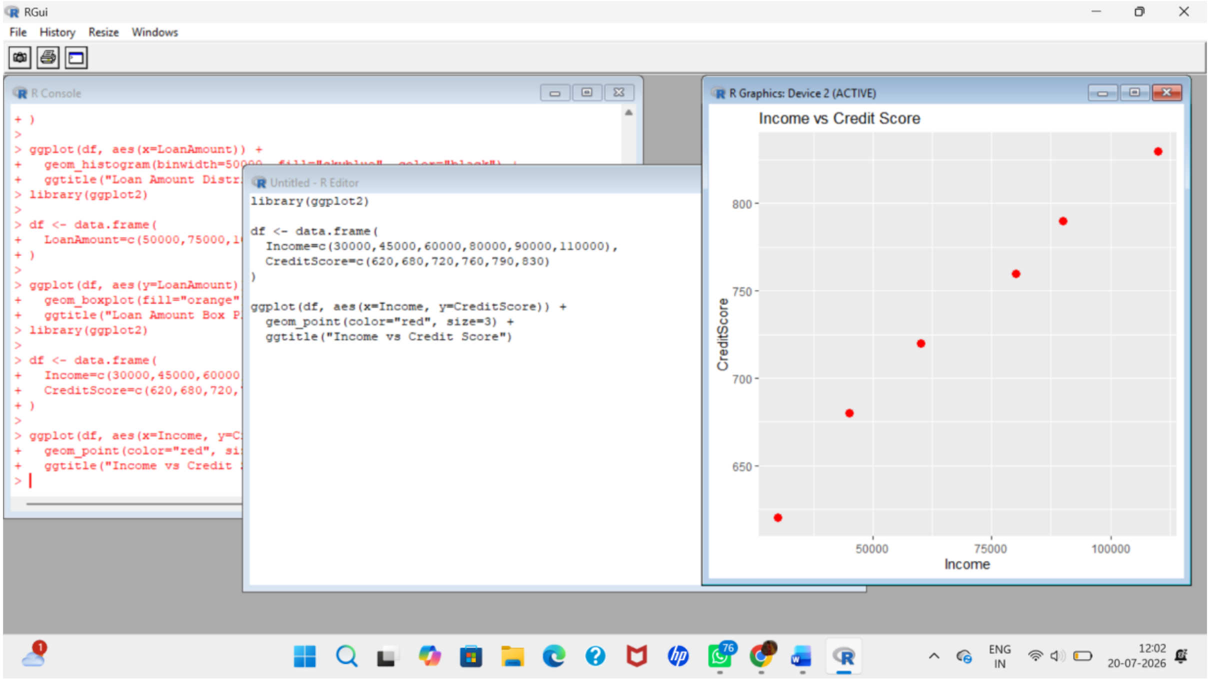
2. Box Plot (Detect Outliers)

```
library(ggplot2)

df <- data.frame(
LoanAmount=c(50000,75000,100000,120000,150000,180000,200000,220000,250000,300000)
)

ggplot(df, aes(y=LoanAmount)) +
  geom_boxplot(fill="orange") +
  ggtitle("Loan Amount Box Plot")
```



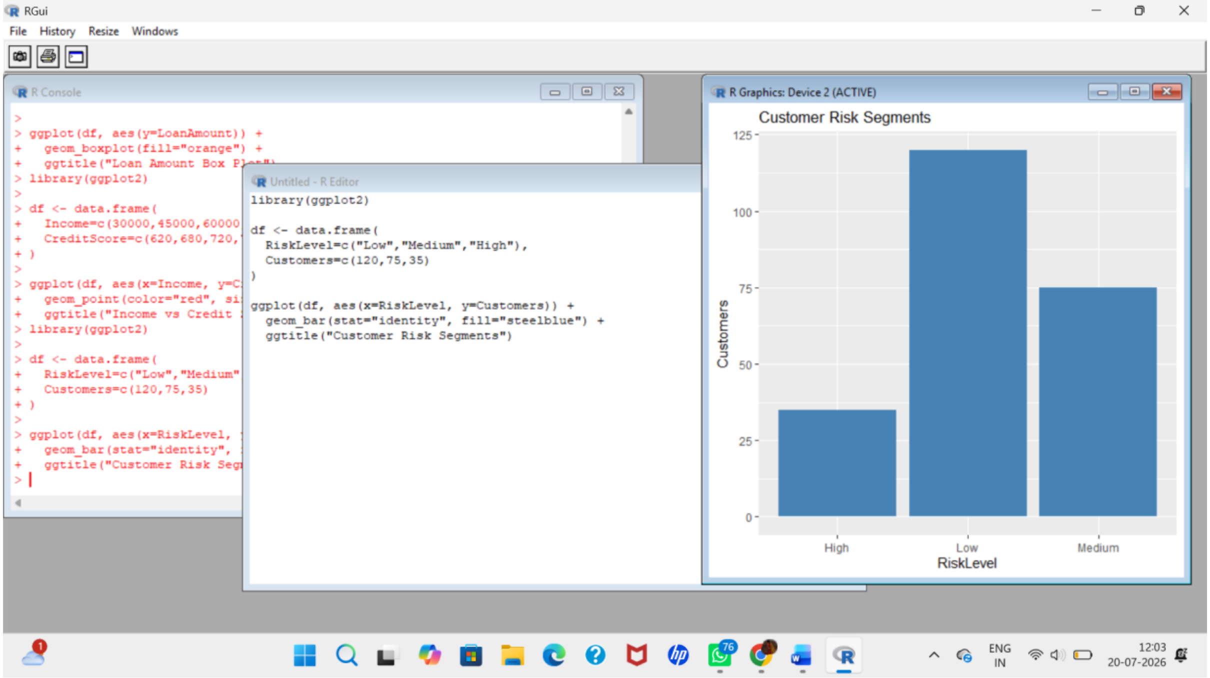
3. Scatter Plot (Income vs Credit Score)

```
library(ggplot2)

df <- data.frame(

  Income=c(30000,45000,60000,80000,90000,110000),

  CreditScore=c(620,680,720,760,790,830)

)

ggplot(df, aes(x=Income, y=CreditScore)) +

  geom_point(color="red", size=3) +

  ggtitle("Income vs Credit Score")
```



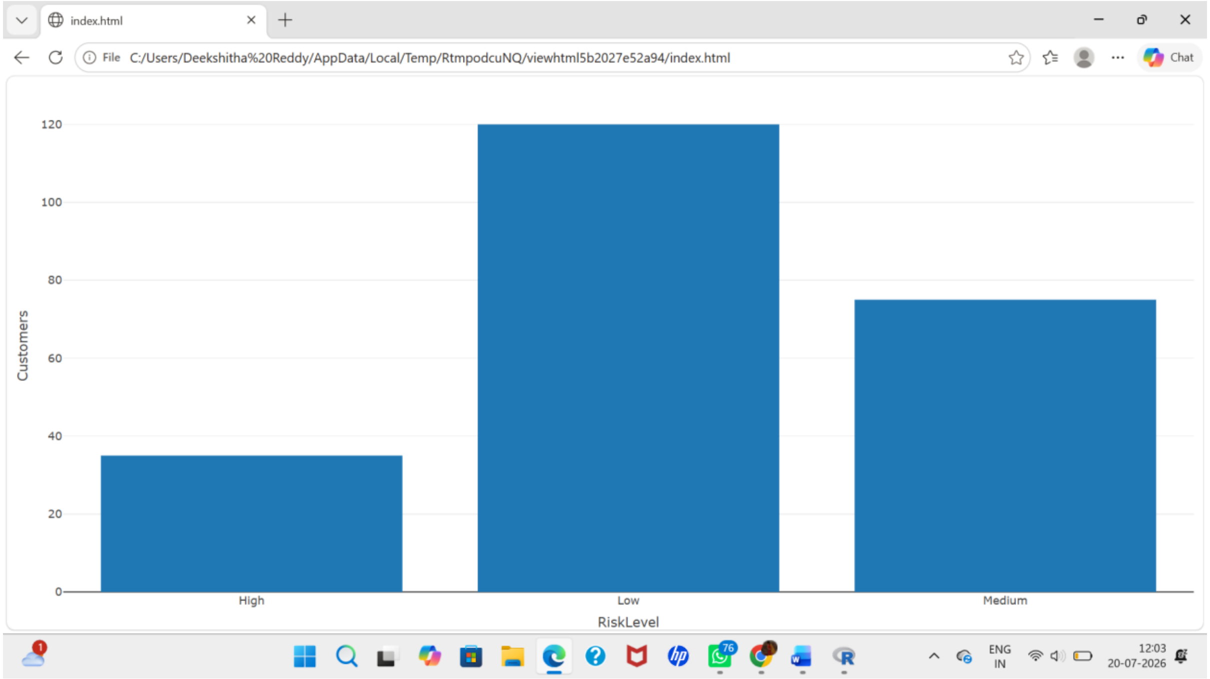
4. Bar Chart (Customer Segments)

```
library(ggplot2)
df <- data.frame(
  RiskLevel=c("Low","Medium","High"),
  Customers=c(120,75,35)
)
ggplot(df, aes(x=RiskLevel, y=Customers)) +
  geom_bar(stat="identity", fill="steelblue") +
  ggtitle("Customer Risk Segments")
```



5. Interactive Bar Chart (Plotly)

```
library(plotly)
df <- data.frame(
  RiskLevel=c("Low","Medium","High"),
  Customers=c(120,75,35)
)
plot_ly(
  df,
  x=~RiskLevel,
  y=~Customers,
  type='bar'
)
```


index.html
File C:/Users/Deekshitha%20Reddy/AppData/Local/Temp/RtmpodcuNQ/viewhtml5b2027e52a94/index.html
Chat
120
100
80
60
40
20
0
Customers
High
Low
Medium
RiskLevel
ENG
IN
12:03
20-07-2026